Power flags (ERC drivers for rails that enter through connectors)

+15V +40V -15V -40V GND

Power input and decoupling

J1 PWR_15V · J2 PWR_HV

C201 47u 25V · C202 47u 25V · C203 100u 63V · C204 100u 63V

U1C U2C U3C U4C U5C U6C NE5532

C211 C213 C215 C217 C219 C221 100n

C212 C214 C216 C218 C220 C222 100n

J1: +/−15 V from PSU (3-pin). J2: +/−40 V from PSU (4-pin, different size so they cannot be swapped).

Master, aux in (U3A), subsonic filter (U6A), bridge phase splitter (U3B)

MASTER 10k A · R41 10k · J6 AUX IN jack · R44 100k · R45 100k · C45 1u film

C46 1u film · R42 22k · R43 22k · R46 10k · C47 47p COG

C51 220n · C52 220n · R61 22k · R62 27k · R63 47k · R64 56k

J11 DEEP/TIGHT sw · U3A SUB NE5532 · U6B U6BD NE5532 · R65 10k 0.1% · R66 10k 0.1% · U3B SUB_INV NE5532

J15 TO POWER AMP

J11 to a DPDT: closing both poles = TIGHT (-40 Hz); open = DEEP (22 Hz). J15 to the power amp board.

A1: bootstrapped instrument input / gain (+1.7 .. +27 dB)

J3 INPUT jack · R1 1k · C1 100p COG · C2 2u2 100V film · R2 1Meg · R3 100k

C3 2u2 63V film · J20 A1 (HVOA-1 socket) · GAIN 50k A · R4 470 · C4 22p COG · R5 2K2

C5 220u 16V · R6 100 · C6 4u7 63V film · R7 100k

RA (R2) bootstrapped by C3 from INN: >1 MOhm input. VR1 rheostat, gain rises CW. C2/C3: film, 100 V.

FX send (U2B) and FX return (A2, HV unity buffer)

J10 SEND LEVEL sw · R31 6k8 · R32 3k3 · R33 100k · U2B SNDO NE5532 · R34 100

C31 10u NP · R35 100k · J4 FX SEND jack · J5 FX RETURN jack · R21 1k · C41 100p COG

C42 2u2 100V film · R22 1Meg · R23 100k · C43 2u2 63V film · J21 A2 (HVOA-1 socket) · R24 1k

R25 100 · C44 4u7 63V film

J5 is a switched jack: its switch lug (pin 2) carries EQ_OUT, so the loop is closed when nothing is plugged in.

Balanced DI / line out (U5)

J12 DI PRE/POST sw · RB1 2k2 · DB1 1N4148 · DB2 1N4148 · RB2 10k · RB3 1k1

J13 PAD sw · U5A HOTO NE5532 · RB5 10k · RB6 10k · U5B COLDO NE5532 · RB7 47

DB3 1N4148 · DB4 1N4148 · CB1 47u 63V · RB8 10 · RB9 100k · RB10 47

DB5 1N4148 · DB6 1N4148 · CB2 47u 63V · RB11 10 · RB12 100k · J8 XLR out (2 hot, 3 cold, 1)

J14 GND LIFT sw

J12: PRE/POST select. J13: closing it = −20 dB pad. J14: ground-lift switch. CB1/CB2 + toward the XLR (phantom).

EQ: Baxandall bass / treble (U1A)

R11 10k · VR2 BASS 50k B · R12 10k · C11 47n · C12 47n · R13 4k7

C13 6n8 · VR3 TREBLE 50k B · C14 6n8 · R14 1k8 · U1A NE5532

CW = boost (wiper toward the input-side end).

EQ: mid, gyrator with 5-way frequency switch (U1B, U2A)

R15 3k3 · U1B EQ_OUT NE5532 · R16 3k3 · VR4 MID 10k B · R17 100k · U2A NE5532

R18 680 · J9 MID FREQ switch · C21 220n · C26 27n · C22 100n · C27 15n

C23 68n · C28 8n2 · C24 33n · C29 4n7 · C25 18n · C30 2n2

J9 to a 2-pole rotary (pole A: MW -> Cs1..5, pole B: GP -> Cg1..5). 250/500/800/1.5k/3k Hz. Short shielded wiring.

Headphone amplifier (U4)

U4 PHONES NE5532 · R71 47 · U4B PHPWR NE5532 · R72 47 · J7 PHONES jack

Phones jack switch contacts are wired to the power amp board (J5 there) to mute the speaker.

PRE clip indicator

R93 447 · D91 1N4148 · R94 447 · D92 1N4148 · C91 1u 50V · R95 220k

D93 1N4740A 10V · R96 10k · R97 10k · Q91 2N5210 · R98 1k5 · J16 PRE CLIP LED

Lights at about +/−11.3 V peak (2.5 dB before the NE5532 stages clip). J16 to a panel LED.

Bipolar (NP) electrolytics use the CP footprint: orientation does not matter for them.
Keep the +/−40 V traces away from the NE5532 area: every HV output enters the 15 V domain AC-coupled.
A1/A2 are HVOA-1 daughterboards (hvoa_module project) plugged into J20/J21.
Pots are PCB-mounted Alpha 16 mm (pin 3 = CW end). Jacks and switches are panel-mounted, wired.

Sheet: /	
File: preamp.kicad_sch	
Title: Bass amp preamp board (HV front end, EQ, I/O)	
Size: A1	Date:
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4	Id: 1/1
	Rev: 1